Arista Access Node KOR-ACC-79 QoS Congestion Event Resolution SOP

Node: KOR-ACC-79 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_5431

Description

The QoS congestion event on the Arista Access node KOR-ACC-79 is caused by the microbursts and interface tail drops due to the misconfigured QoS queues. In an Arista network, QoS is configured using the Modular QoS CLI (MQC). The MQC is a framework that allows network administrators to classify traffic into different classes, apply policies to these classes, and manage the bandwidth allocation for each class. In the Access node, the MQC configuration is used to ensure that high-priority traffic, such as VoIP and video streaming, gets preferential treatment over low-priority traffic, such as HTTP and FTP. However, if the QoS queues are not properly configured, it can lead to microbursts and interface tail drops, resulting in increased latency and packet loss. This can have a devastating impact on mission-critical applications and services. Therefore, it is essential to troubleshoot and resolve QoS congestion events on the Access node promptly.

Pre-requisite Checks

- Ensure that the Koramangala Datacenter is accessible and the remote hands are dispatched to the site.

- Verify that the redundant link status is normal and the network is not experiencing any other issues.

- Backup the running configuration of the Access node before making any changes.

Potential Root Causes

Cause	Probability
Misconfigured QoS queues	High
Insufficient bandwidth allocation for high-priority traffic	Medium
Interface congestion due to high traffic load	Low

Diagnostic Commands & Expected Outputs

```
Command: show traffic interface KOR-ACC-79 | ex bytes, Expected Output: Interface Rx bytes Tx bytes, Description: This command displays the traffic statistics for the interface KOR-ACC-79. The expected output shows the number of bytes received and transmitted on the interface. We can use this command to identify if the interface is experiencing high traffic load, which could be contributing to the QoS congestion event.

Command: show qos queue KOR-ACC-79 | ex queue, Expected Output: Queue Max-Size Current-Size, Description: This command displays the QoS queue configuration for the Access node. The expected output shows the maximum and current sizes of the queue. We can use this command to
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verify if the QoS queues are properly configured and if they are
experiencing any issues.
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Resolution Steps

Step 1: Step: Verify QoS queue configuration, Command: show qos queue KOR-ACC-79 | ex queue, Description: Verify that the QoS queues are properly configured and that they are not experiencing any issues. If the configuration is incorrect, correct it and verify that the queues are functioning properly.

Step 2: Step: Adjust bandwidth allocation for high-priority traffic, Command: bandwidth KOR-ACC-79 priority high 1000, Description: Adjust the bandwidth allocation for high-priority traffic to ensure that it gets sufficient bandwidth to prevent QoS congestion events.

Step 3: Step: Clear interface statistics, Command: clear traffic interface KOR-ACC-79, Description: Clear the interface statistics to ensure that the interface is functioning properly and that there are no issues with the traffic load.

Post-Check Verification

```
Command: show traffic interface KOR-ACC-79 | ex bytes, Expected
Output: Interface Rx bytes Tx bytes, Description: Verify that the
traffic statistics for the interface KOR-ACC-79 are normal and that
the interface is functioning properly.

Command: show qos queue KOR-ACC-79 | ex queue, Expected Output: Queue
Max-Size Current-Size, Description: Verify that the QoS queue
configuration for the Access node is correct and that the queues are
functioning properly.
```

Roll-back Procedures

- Step: Restore the original configuration, Command: copy running-config startup-config, Description: Restore the original configuration of the Access node to resolve the QoS congestion event.